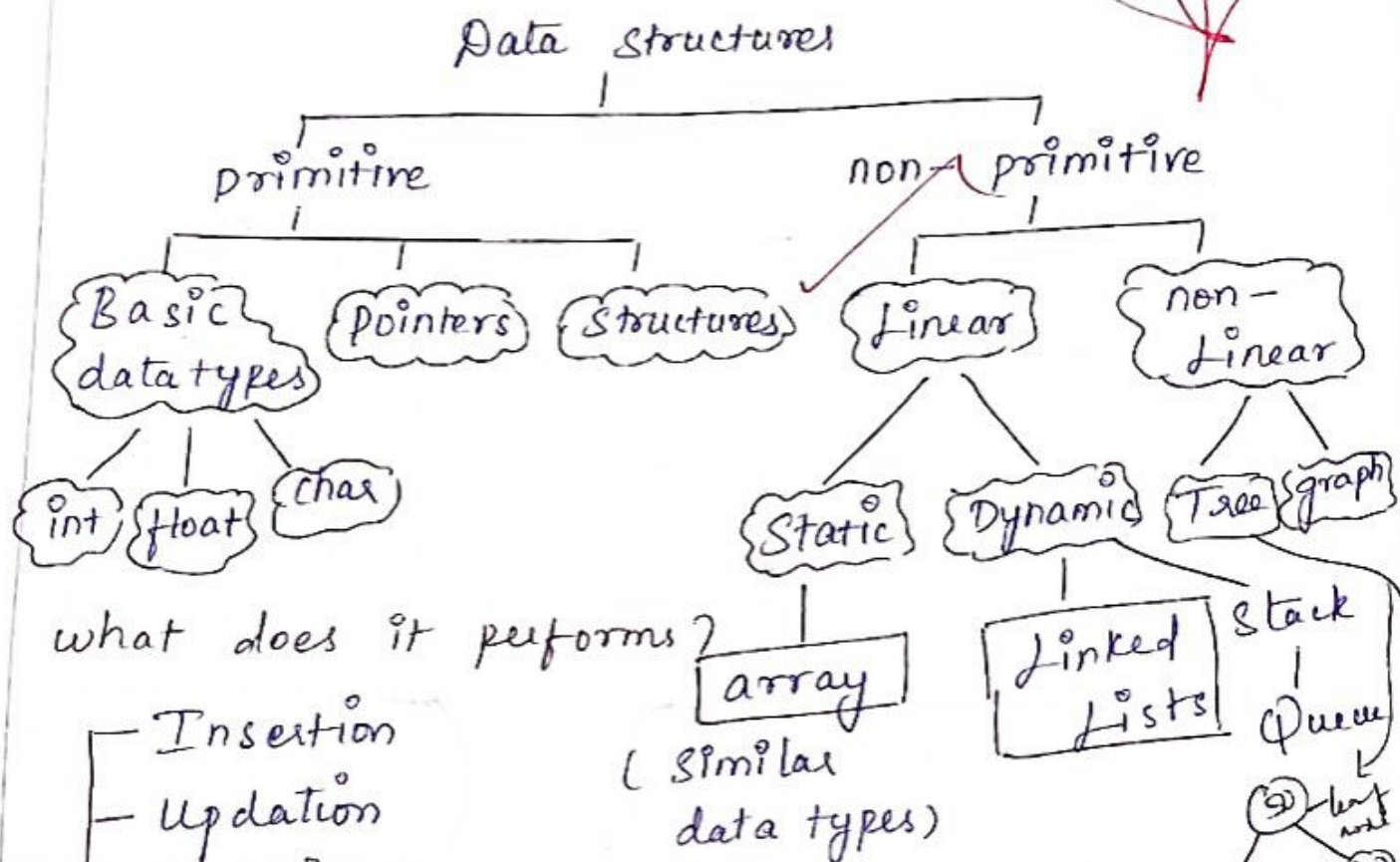


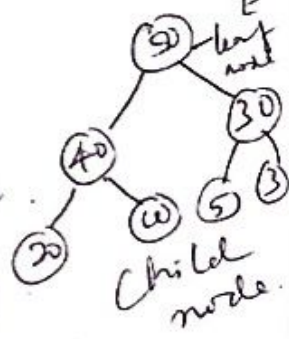
① Data Structures :-

* It is a method of organising or arranging the data in the primary memory so that the users can easily access and stores the data

Classification of data structures



- * Organizes the data.
- * Performs Searching
- * Sorting the data.
- * Easy to store and Access.



For Example :-

array =

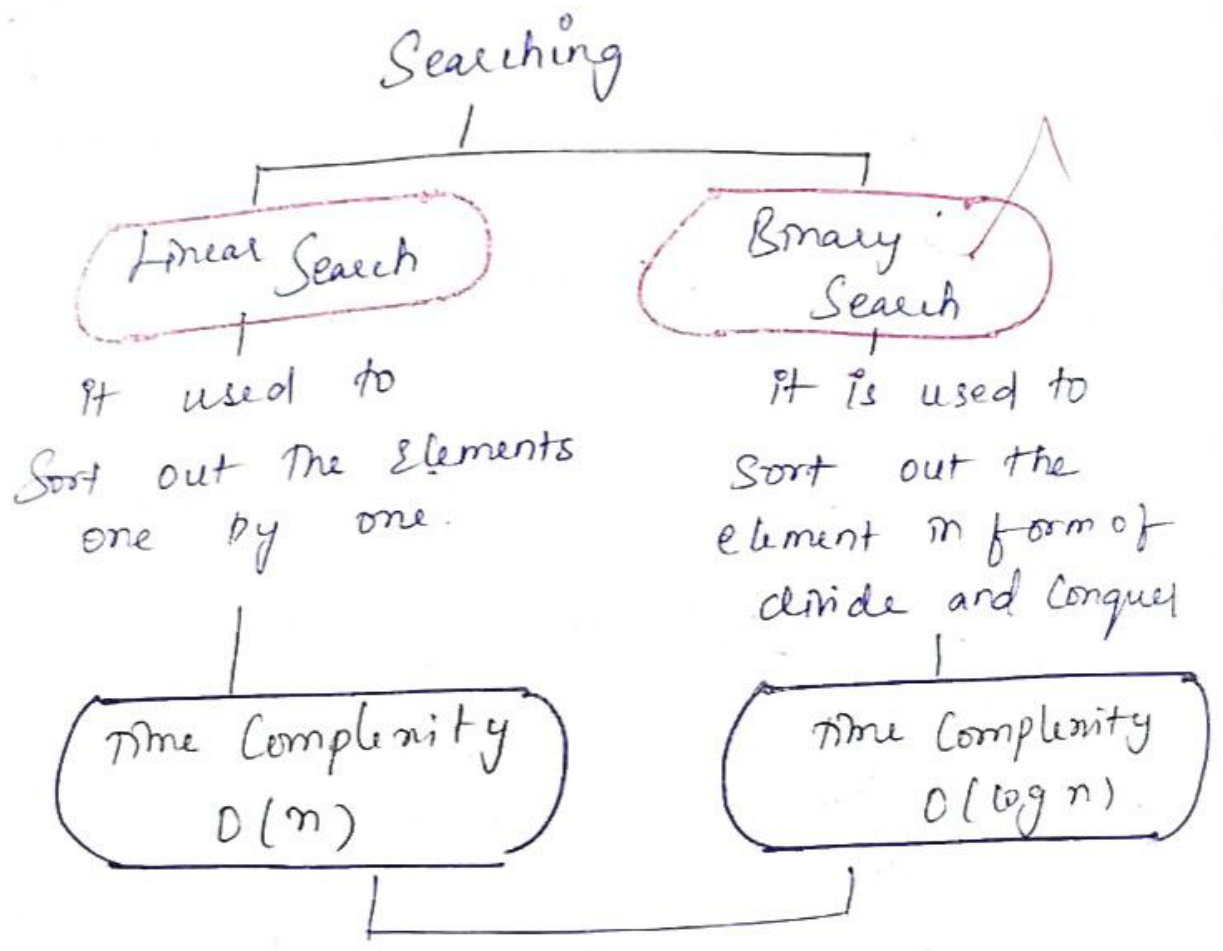
0	1	2	3	4
15	20	70	80	90

Example key

- * It comes under the Linear data structure which performs static form of storing the arrays like no. of vehicles, Speed of vehicles.
- * Efficiently in forms of Tree structures.

Linear Search (Searching):

(2)



Efficiency

- * Compared to Both Binary and Linear search Binary Search is Efficient and faster than Linear Search.

Example for Linear Search program

```
#include <stdio.h>
```

```
int main() {
```

```
    int i, n, key = 30, found = 1;  
    key = 30
```

```
    scanf("%d %d", &n);
```

```
    for (i = 0; i < n; i++)
```

```
        found = 1;
```

```
    if ("Element is found:")
```

```
        break;
```

```
    if ("Element not found:")
```

```
        return 0;
```

```
}
```

} shows slower
than Binary
Search

Binary Search Example to show it faster search

int (i, n)

get an array element of
a[50]

key = 30

found = 1

if (i = 0; i < n; i++) (i+1)

Element found.

if (i == 1)

Element not found

return.

③

Library Management System
Title, Author, Availability
Status.

Library management using data structures

Organisation
using array
shows no. of Books
etc.

Access methods
using Stack,
Queue,
Linked List

Efficiency
Based upon
Time
complexity

Stores the
data in the form
of Arrays.

- Titles
- Authors
- Availability of Books

A[50] =

0	1	2	3	4
15	30	70	95	100

{ Key = 30 index = 1
Array shows the
position & index.

Performances in
data structures

- Insert
- Update
- Delete
- Retrieve

15	30	70	90	100
----	----	----	----	-----

Insert 50 in place
of 70

① perform sorting

15	30	70	90	100
----	----	----	----	-----

Move one by one

15	30	50	90	100
----	----	----	----	-----

Struct Box

```
{ int bid;  
  char t[20], a[20];  
  struct Book * next;  
  }  
head
```

1	DS	Lia
---	----	-----

2	CS	Arjun
---	----	-------

3	EVS	Prity
---	-----	-------

4	Maths	Shruti
---	-------	--------

4 Books
for management
of Library